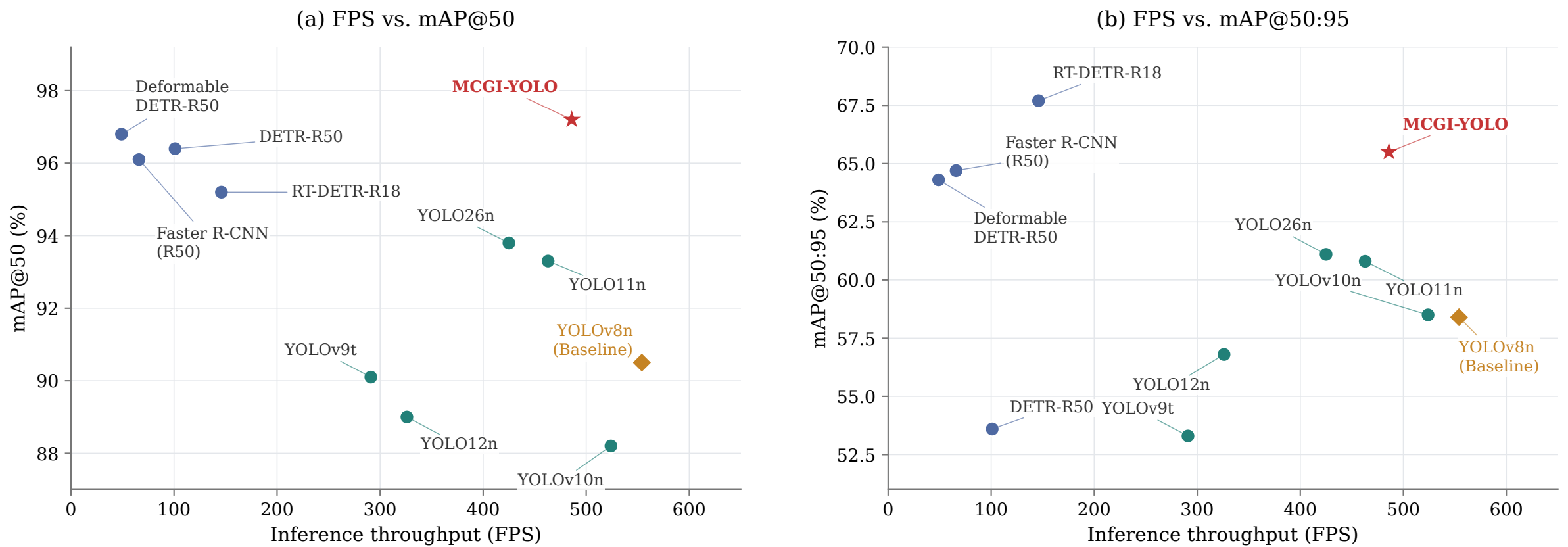
(a) FPS vs. mAP@50
mAP@50 (%)
Inference throughput (FPS)
Deformable DETR-R50
DETR-R50
Faster R-CNN (R50)
RT-DETR-R18
MCGI-YOLO
YOLO26n
YOLO11n
YOLOv8n (Baseline)
YOLOv9t
YOLO12n
YOLOv10n
(b) FPS vs. mAP@50:95
mAP@50:95 (%)
Inference throughput (FPS)
RT-DETR-R18
Faster R-CNN (R50)
Deformable DETR-R50
MCGI-YOLO
YOLO26n
YOLOv10n
YOLO11n
YOLOv8n (Baseline)
YOLO12n
DETR-R50
YOLOv9t
RTX 3090 | FP32 | Batch size 1 | Input 640 x 640